Activity: Poisson assumptions and diagnostics

Biochemistry PhD students

We have information on 915 biochemistry PhD students, with variables including:

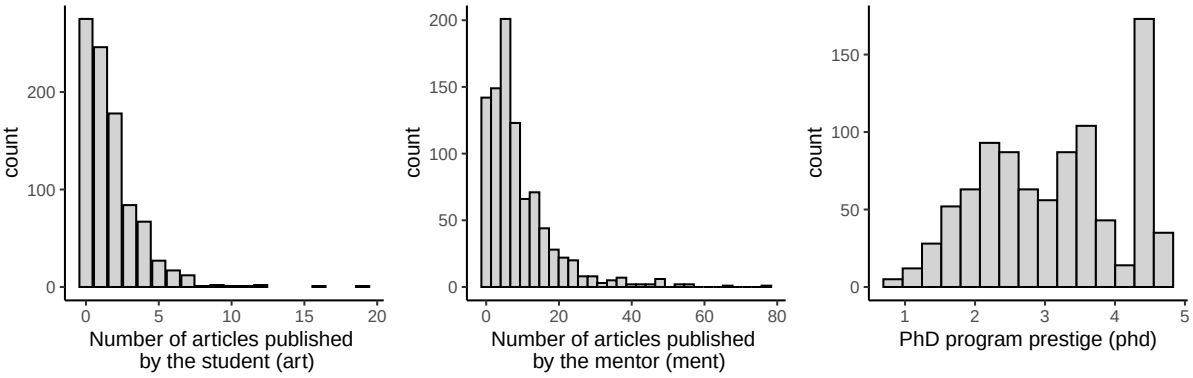
- `art`: articles published by the student in last three years of Ph.D.
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program (a numeric score from 0 to 5)
- `ment`: articles published by their mentor in last three years

We are interested in modeling the number of articles published by the student in the last three years of their PhD, and so we fit a Poisson regression model.

Data and EDA

Characteristic	**N = 915**[1]
art	1 (0, 2)
phd	3.15 (2.26, 3.92)
ment	6 (3, 12)
kid5	
0	599 (65%)
1	195 (21%)
2	105 (11%)
3	16 (1.7%)

[1]Median (Q1, Q3); n (%)



Model

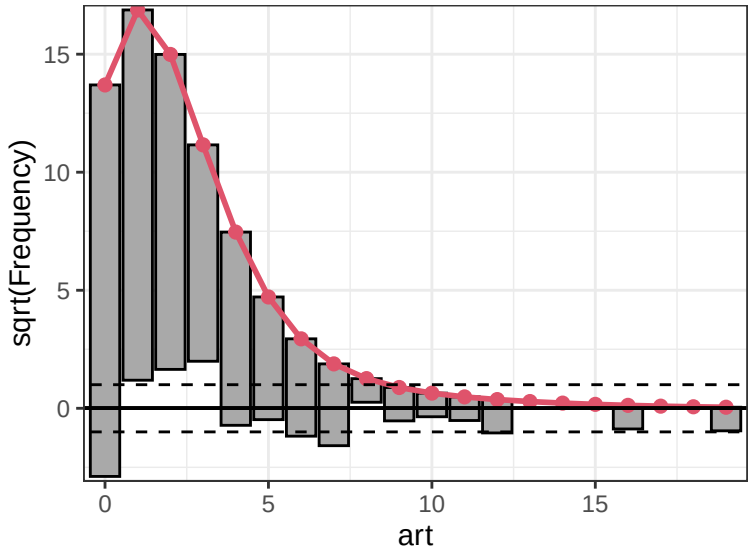
```
m_art <- glm(art ~ phd + ment + kid5,
             family = poisson, data = articles)
```

Characteristic	IRR	95% CI	p-value
phd	1.01	0.96, 1.06	0.8
ment	1.03	1.02, 1.03	<0.001
kid5	0.91	0.84, 0.97	0.005

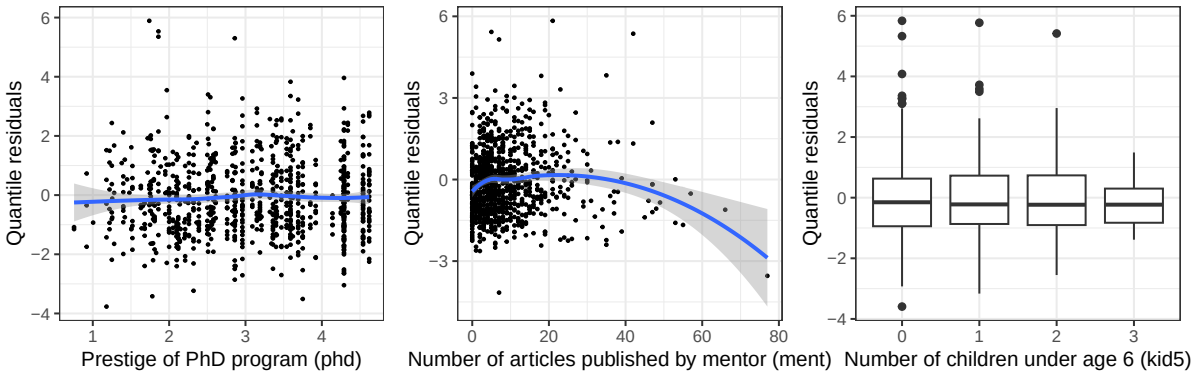
Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio

(The *Incidence Rate Ratio* is just e^{β}, i.e. the multiplicative change in the estimated mean response).

Rootogram

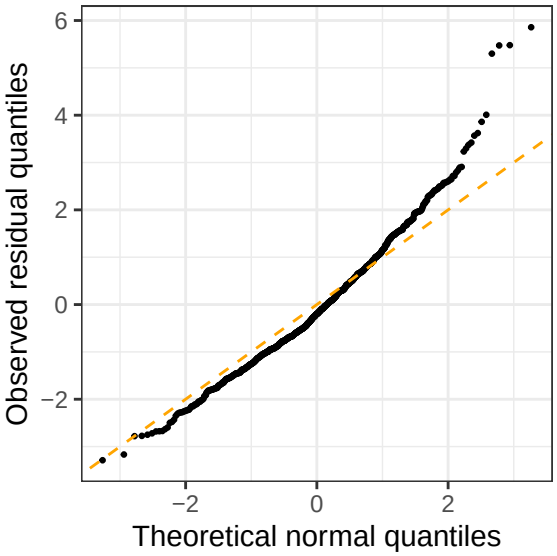


(Randomized) quantile residual plots

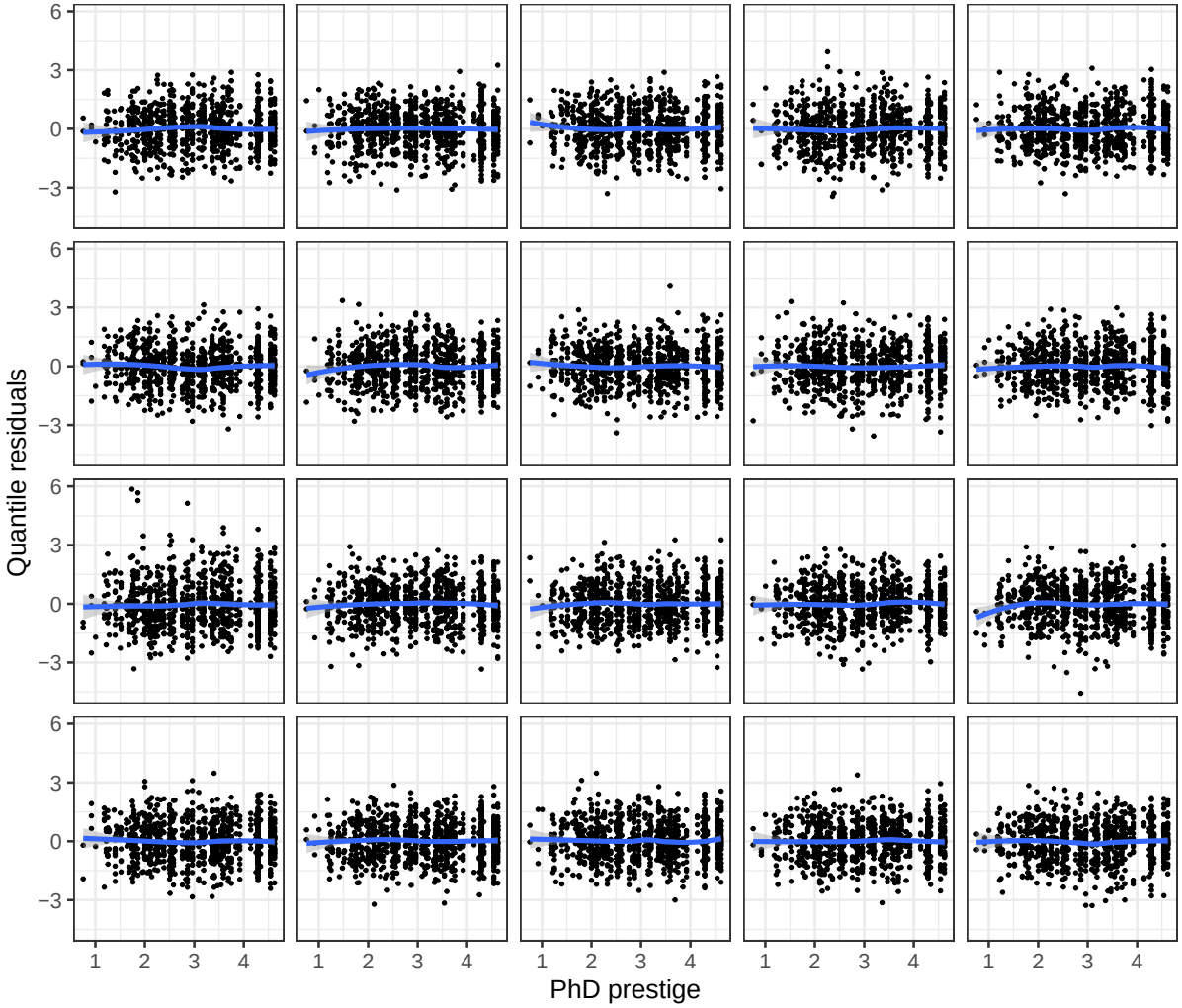


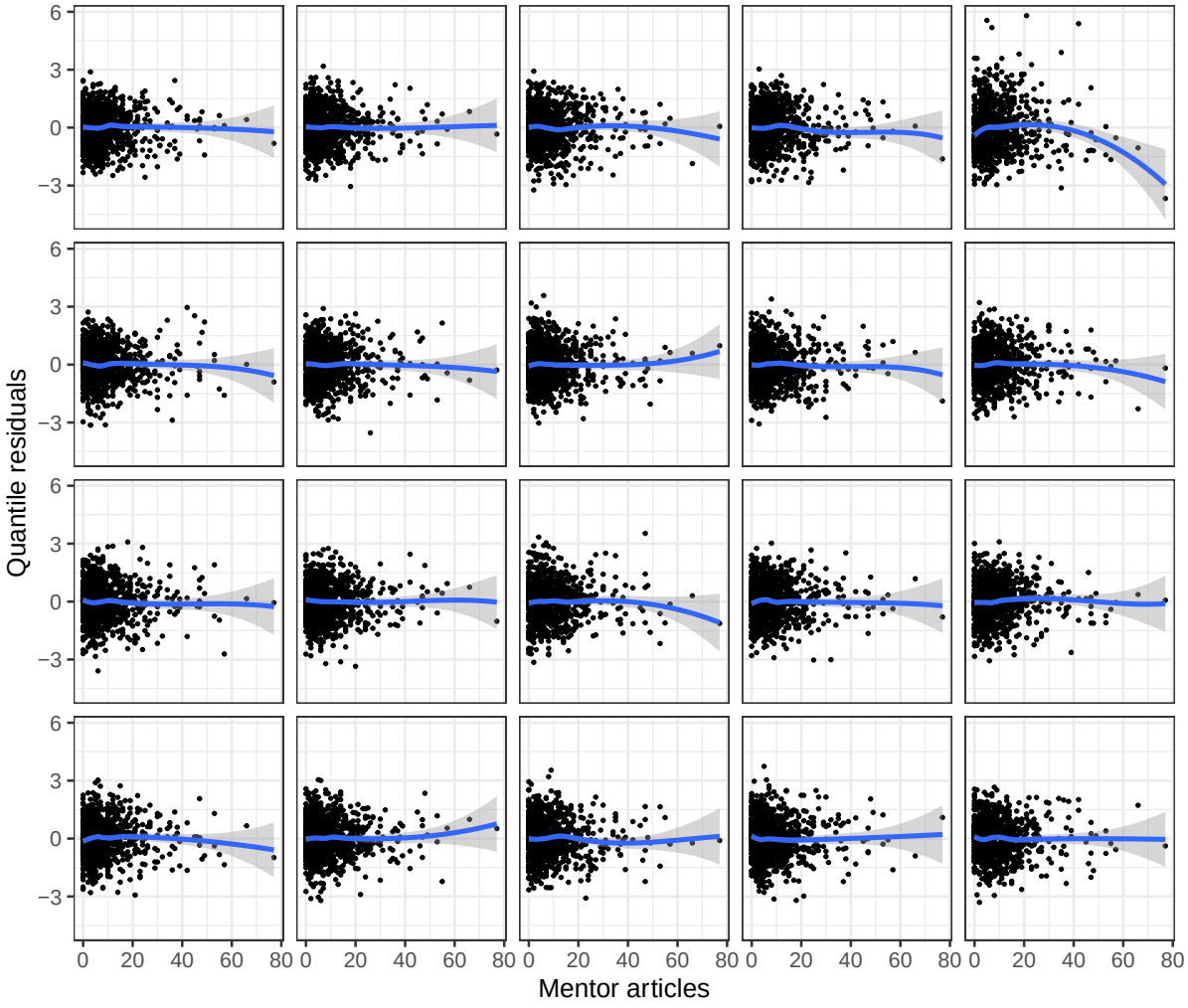
QQ plot of quantile residuals

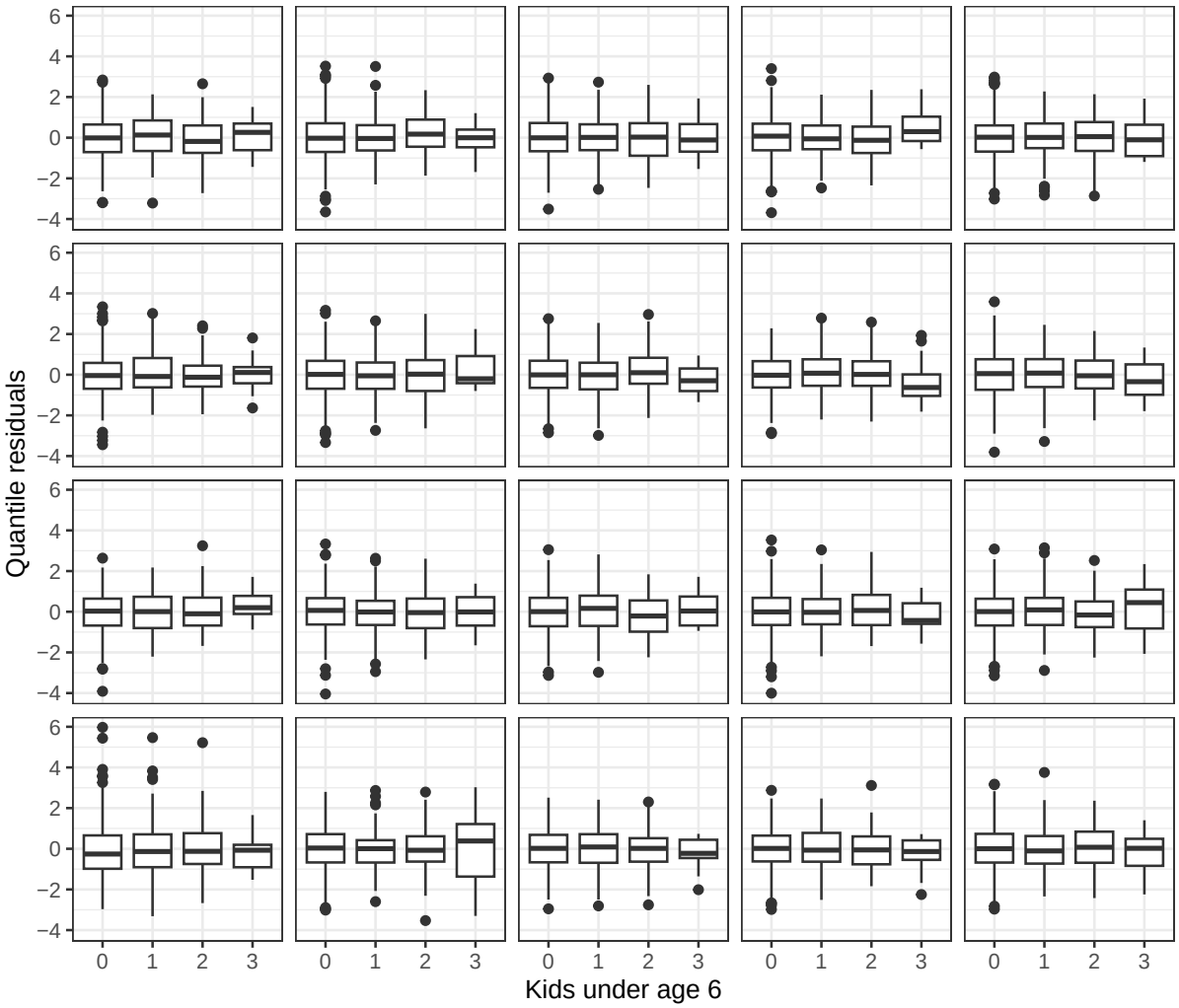
If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$. The dashed orange line is the line of slope 1 and intercept 0.



Plot lineups




Quantile residuals
6
3
0
−3
0
20
40
60
80
Mentor articles



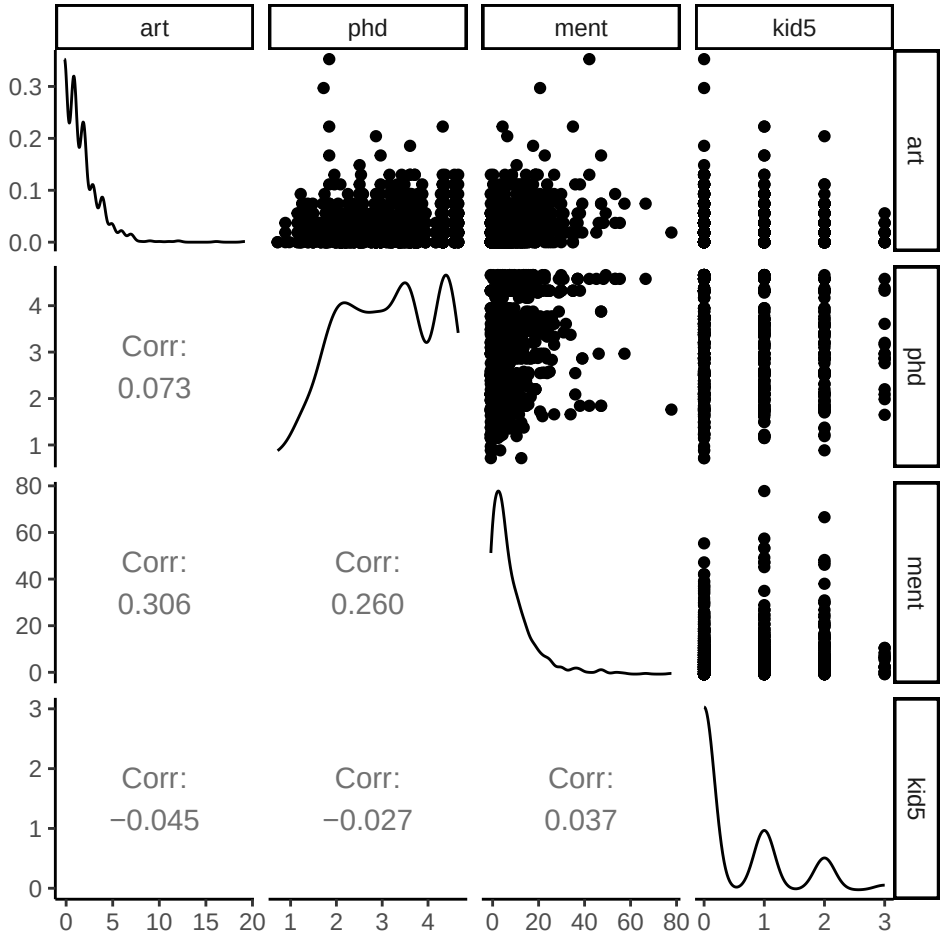
Cook's distance

The following table shows the first few rows of the data with their associated Cook's distances, ordered by descending Cook's distance. (The numbers next to the table are just the row numbers in the original data).

	art	phd	ment	kid5	cooks_distance
328	1	1.78	77	1	0.5587585
915	19	1.86	42	0	0.4113048
914	16	1.74	21	0	0.1478051
911	11	2.86	7	2	0.0905633
912	12	4.29	35	1	0.0638557

	art	phd	ment	kid5	cooks_distance
913	12	1.86	5	1	0.0625560
909	9	1.86	47	1	0.0481382
669	2	4.54	55	0	0.0351612
803	4	4.54	66	2	0.0243971
883	6	2.51	11	2	0.0199222

Correlation and VIFs




```
car::vif(m_art)
```

```
     phd     ment     kid5
1.057959 1.071738 1.015631
```